

4	(a)	direction of acceleration opposite direction of displacement from equilibrium position (or directed to fixed point) B1 (magnitude of) <u>acceleration proportional</u> to (magnitude of) <u>displacement from equilibrium position</u> B1 (max 1 mark if missing from entire definition)	
	(b)		
		(i) $a = \omega^2 x_0 = (2\pi \times 1.5)^2 (0.034) = 3.0201 \text{ m s}^{-2}$ M1 $F = ma = 1.2 \times 3.0201 = 3.6 \text{ or } 3.62 \text{ m s}^{-2} \text{ (2 or 3 sf)}$ A1	
		(ii) $v_0 = \omega x_0 = (2\pi \times 1.5) (0.034) = 0.32044 \text{ m s}^{-1}$ C1 $E_k = \frac{1}{2} mv^2 = \frac{1}{2} (1.2) (0.32044)^2 = 0.062 \text{ or } 0.0616 \text{ J}$ A1	
	(c)	<u>no oscillations</u> of readings B1 returns to equilibrium position (or actual measurement) in the shortest time B1	

5	(a)	The electric current I (conventional current) at a particular point is the rate of flow of	
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